

KILOMINX

12 SIDES · 20 CORNERS · ONE SECRET

SOLUTION GUIDE



UNLOCK THE SECRET!

STAGE 1:

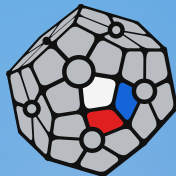
GET TO KNOW YOUR KILOMINX

DEFINITIONS OF KILOMINX PIECES

THE PARTS:

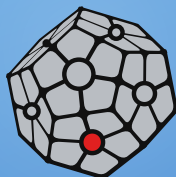
CORNER PIECES

PIECES WITH THREE (3) COLORS. THERE ARE TWENTY (20) CORNER PIECES — these are the **ONLY** pieces you move. The whole puzzle is corners!



CENTER PIECES

PIECES WITH ONE (1) COLOR. THERE ARE TWELVE (12), ONE ON EACH FACE. THEY **DO NOT MOVE** and show the color of their face.



A kilominx is a megaminx with **no edge pieces** — just corners and centers. It has **12 faces** and **12 colors**; each color is on exactly one face. **WHITE** and **GRAY** are on opposite faces — you start with the white face on top and finish with the gray one on the bottom. (Use the face colors of YOUR puzzle; the steps are the same.)

EACH FACE OF THE KILOMINX IS REPRESENTED BY A LETTER

- U** = **Up Face** — the top face
- F** = **Front Face** — the face facing you, below the top
- R** = **Right Face** — the face just right of the front
- L** = **Left Face** — the face just left of the front
- D** = **Down Face** — the face below and between F and R

A letter with an "i" after it means an **inverted** (counter-clockwise) turn of that face. A letter alone means a clockwise turn (as you look at that face). Every turn is **one click** — one fifth of the way around.

VERY IMPORTANT

The pictures show exactly which face moves and which way. Hold your kilominx to match the picture before each sequence. Dark gray on a picture means **the color there does not matter**.



STAGE 2:

SOLVE THE WHITE CORNERS

HOLDING YOUR KILOMINX: Hold your kilominx with the **WHITE** center on top (**U**). You will fill the five corners around the white face, each with white on top and its two side colors matching the side centers.

TIPS:

- Find a white corner in the bottom of the puzzle. Park it directly **below the slot it belongs in** by turning the bottom faces.
- Then repeat the **Righty** move below — **(R) i (D) i (R) (D)** — until the corner pops up into place with white on top. Repeat 1, 2, or 3 times.
- Turning the bottom faces never disturbs corners already finished on top, so work one corner at a time.



Ri



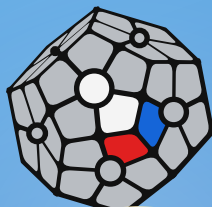
Di



R



D



YOUR GOAL

STAGE 3:

SOLVE THE UPPER MIDDLE CORNERS

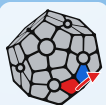
HOLDING YOUR KILOMINX: Keep white on top. Now fill the ring of five corners just below the white ones — each has two side colors and no white and no gray.

TIPS:

- These corners are already in the middle — find one that is in the wrong place (or twisted), and use the same **Righty** move to kick it out and bring the right one in.
- Park the corner you want directly below its slot, then repeat **Righty** until it seats. Same move as the white corners, one row lower.



Ri



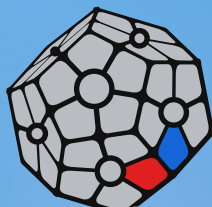
Di



R



D



YOUR GOAL

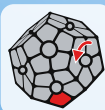
STAGE 4:

SOLVE THE LOWER MIDDLE CORNERS

HOLDING YOUR KILOMINX: Still white on top. Fill the next ring down — the five corners that sit just above the gray face. None of these has white; none has gray yet on top.

TIPS:

- Exactly like before: park the corner below its slot and repeat **Righty** until it drops in.
- After this stage only the five **GRAY** corners on the bottom are left. Turn the whole puzzle over so gray is on top for the last two pages.



R₁



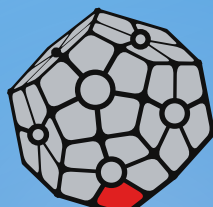
D₁



R



D



YOUR GOAL

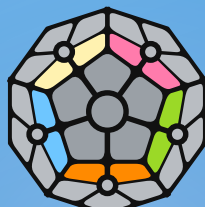
STAGE 5:

LAST LAYER: MOVE THE CORNERS HOME

HOLDING YOUR KILOMINX: Turn the puzzle over so the **GRAY** center is on top (**U**). First get every gray corner into the right CORNER (ignore which way it is twisted for now).

TIPS:

- Hold the puzzle so one corner that is already in the right place is at the back-right. Then do the **Corner Cycle** below; it spins the other three corners around.
- Turn the gray top to look at the result, and repeat the Corner Cycle until all five corners are home. They may still be twisted — that is the next page.



R₁



BR₁



R



BR



R₁



F₁



R

CORNERS IN PLACE



BR₁



R₁



BR



F



R

STAGE 6:

LAST LAYER: TWIST THE CORNERS

HOLDING YOUR KILOMINIX: Still **GRAY** on top. Every corner is in the right place — now twist each one so gray faces up. This uses the **Righty** move you already know.

TIPS:

- Look at the front-right corner. Repeat **Righty** until its gray sticker faces up. The lower part will look scrambled — that is normal, it fixes itself.
- Now turn **ONLY** the gray top face to bring the next un-twisted corner to the front-right, and repeat. Do not turn anything else.
- When the last corner twists up, a final turn of the gray face snaps the whole puzzle solved. You did it!



SOLVED!



Ri



Di



R



D

NOW YOU KNOW THE SECRET...

FUN FACTS!

A kilominx is sometimes called a "**corners-only megaminx**." Take a megaminx, throw away every edge piece, and this is what is left.

It still has **twenty corner pieces** in a dozen colors — plenty of ways to scramble, but always the same secret to solve.

Every solve in this booklet was checked, move by move, on a computer model of the puzzle — so if you follow the pictures exactly, it always works.

The method you just learned is the same idea as the megaminx and your cube: build the white layer, drop in the middle corners with the Righty move, then finish the last layer in two small steps. Different puzzle — same secret.



Made for a young puzzler who already unlocked the cube. Happy twisting!